

PROJECT PROPOSAL

ICT 222-2

Smart Public Transportation Information System

Group Number: 11

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Table of Contents

1. Introduction	
1.1 Project Title.....	1
1.2 Project Description	1
2. Background and Motivation	
2.1 Background of the Problem	2
2.2 Existing Systems and Related Work	2
2.3 State of Art.....	3
2.4 Motivation.....	3
3. Problem Statement and Proposed Solution	4
3.1 Problem Statement.....	4
3.2 Importance of the Problem	4
3.3 Proposed Solution.....	4
3.3.1 Use case diagram.....	5
3.3.2 System Architecture.....	6
4. Aim and Objectives	7
4.1 Aim	7
4.2 Objectives	7
5. Methodology	8
5.1 Software Development Methodology	8
5.2 Technology Stack.....	8
5.3 Functional Requirements	9
5.4 Non-Functional Requirements.....	11
6. Resource Requirements and Budget	13
6.1 Hardware Requirements	13
6.1.1 Passenger Mobile Device Requirements	13
6.1.2 Driver Device Requirements	14
6.1.3 Conductor Device Requirements.....	14
6.1.4 Administrator System Requirements	14
6.2 Software Requirements.....	15
6.2.1 Mobile Application Software Requirements	15
6.2.2 Admin Panel Software Requirements.....	15
6.2.3 Development Environment Requirements	16
7. Project Time Plan	17
8. References	18
9. Signature of the Team Members	19

1. Introduction

1.1 Project Title

Smart Public Transportation Information System

1.2 Project Description

Bus passengers often face long waiting times because there is no proper system to track bus movements in real time. Similarly, train passengers also experience inconvenience due to the lack of easily accessible information about train schedules, delays, and service updates. In many situations, passengers must rely on printed timetables, station announcements, or informal sources that may not provide reliable or updated information.

Another major issue is the absence of a centralized digital platform that combines both bus and train transportation services into a single system. This makes travel planning inefficient and inconvenient for daily commuters.

Although modern smart transportation technologies are widely used in many countries, such systems are still limited in local public transportation services. Therefore, there is a need for a smart and user-friendly mobile application that provides real-time tracking, route information, schedules, and transport updates for both buses and trains in the Badulla District.

The Smart Public Transportation System is a Mobile application designed to improve the efficiency and reliability of public transport services in the Badulla District, Sri Lanka. The system focuses on integrating both bus and train transportation services into a single platform, allowing users to easily access real-time travel information

2. Background and Motivation

2.1 Background of the Problem

At present, public transportation in the region mainly operates using manual scheduling methods, where passengers depend on physical timetables and waiting to catch transportation service. This often results in uncertainty regarding arrival times, delays, and inefficient travel planning. There is also no centralized digital platform that connects passengers with drivers and conductors to provide live transport updates.

2.2 Existing Systems and Related Works

There are several Mobile Applications that provide the services separately but there is no any system that combine both buses and trains,

Table 1: Existing Systems

Existing System	Features	Limitations
LMT – GO: Lanka Metro Transit	Track Colombo metro buses live, plan journeys and pay fares.	No trains integration.
Highway Bus Sri Lanka	Discover Highway Bus Schedules, Routes, Contact numbers and Bus.	Only shows Bus details and not for booking.
LankaGo	Sri Lanka Bus Schedules. Currently: Mathugama area.	Only for Mathugama Area.
SP BUSES	Live Bus Tracking, nearby buses and Time Table in southern province Buses.	Only for southern province.
RDMNS.LK	provide lives updates and alerts Sri Lanka.	No buses integration.
Trains – Sri Lanka	Fastest and Smartest Train Schedules, Train/Station Times Tables.	Only shows tarin schedules. /No buses integration.
Pravesha	Train Ticket booking system	Not real time tracking system.

2.3 State of the Art

The current state of public transportation systems globally shows a strong shift towards smart and technology-driven solutions. Many developed countries and urban cities have implemented Intelligent Transportation Systems (ITS) that enhance the efficiency, reliability, and user experience of public transport services.

The proposed Smart Public Transportation System aims to address these limitations by developing a mobile application that combines both bus and train transportation services into a single platform. The system will provide real-time tracking, schedules, route details, and transport updates to improve travel planning and passenger convenience within the Badulla District.

2.4 Motivation

The motivation behind developing the Smart Public Transportation System comes from the daily difficulties faced by passengers using bus and train services in the Badulla District, Sri Lanka. Commuters often experience problems such as long waiting times, lack of real-time transport information, and difficulty in planning journeys efficiently due to the absence of a centralized digital platform.

Currently, there are only limited transportation applications available, and most of them focus on either buses or trains separately. According to our observations, there is no existing local system that effectively combines both bus and train transportation services into a single platform. This creates inconvenience for passengers who use multiple modes of transport during their daily travel.

As ICT students, we are motivated to apply our knowledge in mobile application development, databases, GPS integration, and real-time systems to create a solution for a real-world transportation problem. This project also provides valuable practical experience while contributing towards the improvement of smart public transportation services in the Badulla District.

3. Problem

3.1 Problem Statement

Public transportation in the Badulla District, Sri Lanka, mainly depends on buses and trains for daily commuting. However, passengers frequently experience difficulties due to the lack of accurate and real-time transport information. Most transport services still operate using traditional methods, where passengers have limited access to live updates regarding vehicle locations, arrival times, delays, and route information.

Bus passengers often face long waiting times because there is no proper system to track bus movements in real time. Similarly, train passengers also experience inconvenience due to the lack of easily accessible information about train schedules, delays, and service updates. In many situations, passengers must rely on printed timetables, station announcements, or informal sources that may not provide reliable or updated information.

Another major issue is the absence of a centralized digital platform that combines both bus and train transportation services into a single system. This makes travel planning inefficient and inconvenient for daily commuters.

3.2 Importance of The Problem

The lack of a smart and centralized public transportation system creates significant inconvenience for daily commuters in the Badulla District. Passengers often face delays, uncertainty in arrival times, and difficulties in planning their journeys due to the absence of real-time transport information. Addressing this problem is important to improve passenger convenience, reduce waiting time, and enhance the overall efficiency and reliability of public transportation services.

3.3 Proposed Solution

To address the identified transportation issues, the proposed solution is to develop a Flutter-based Smart Public Transportation System for the Badulla District. The mobile application will integrate both bus and train transportation services into a single platform, allowing passengers to access real-time tracking, schedules, route details, and transport updates conveniently through their smartphones. The system aims to improve travel planning, reduce passenger waiting time, and provide a more efficient and user-friendly public transportation experience.

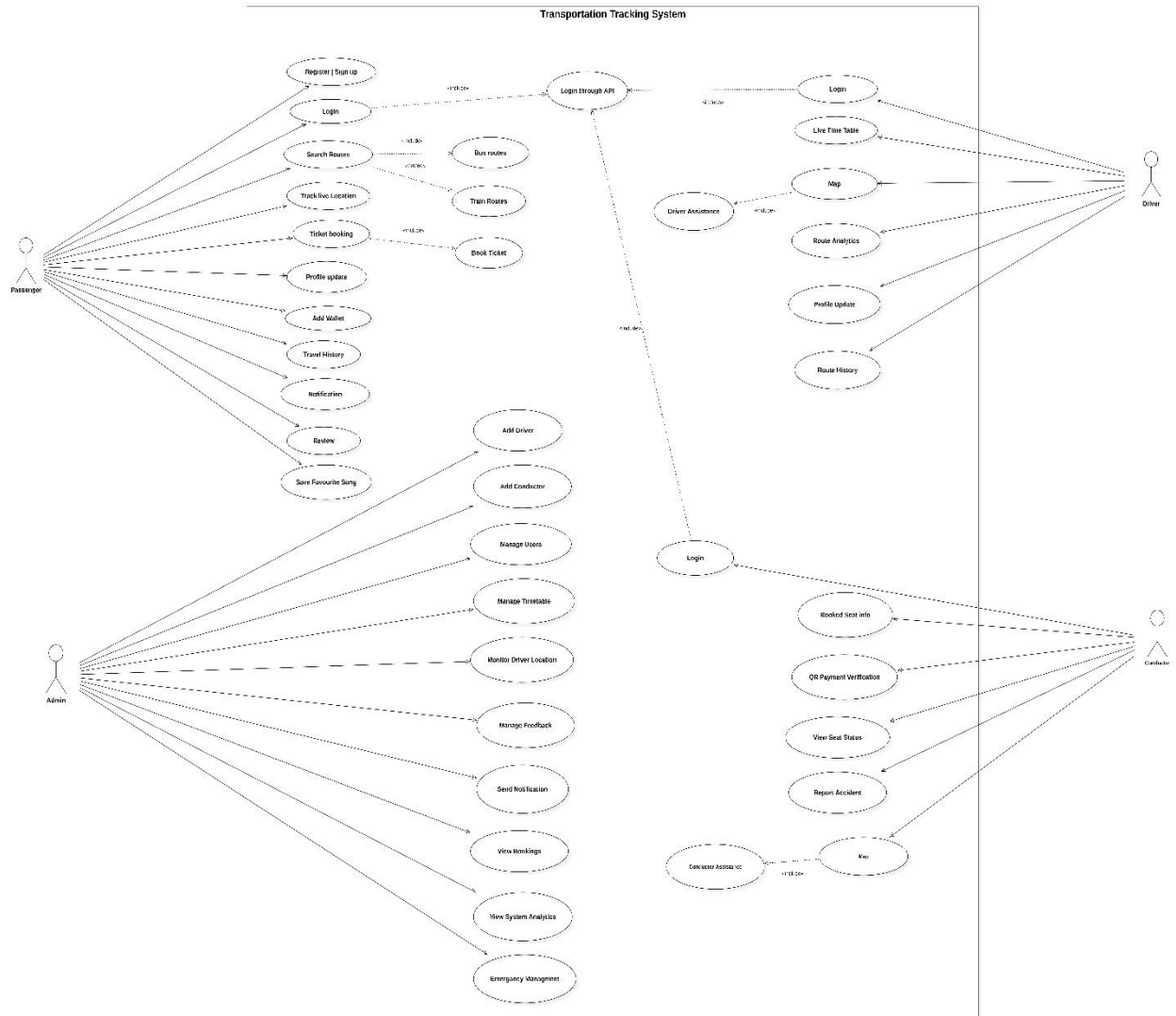


Figure 1: Use Case Diagram

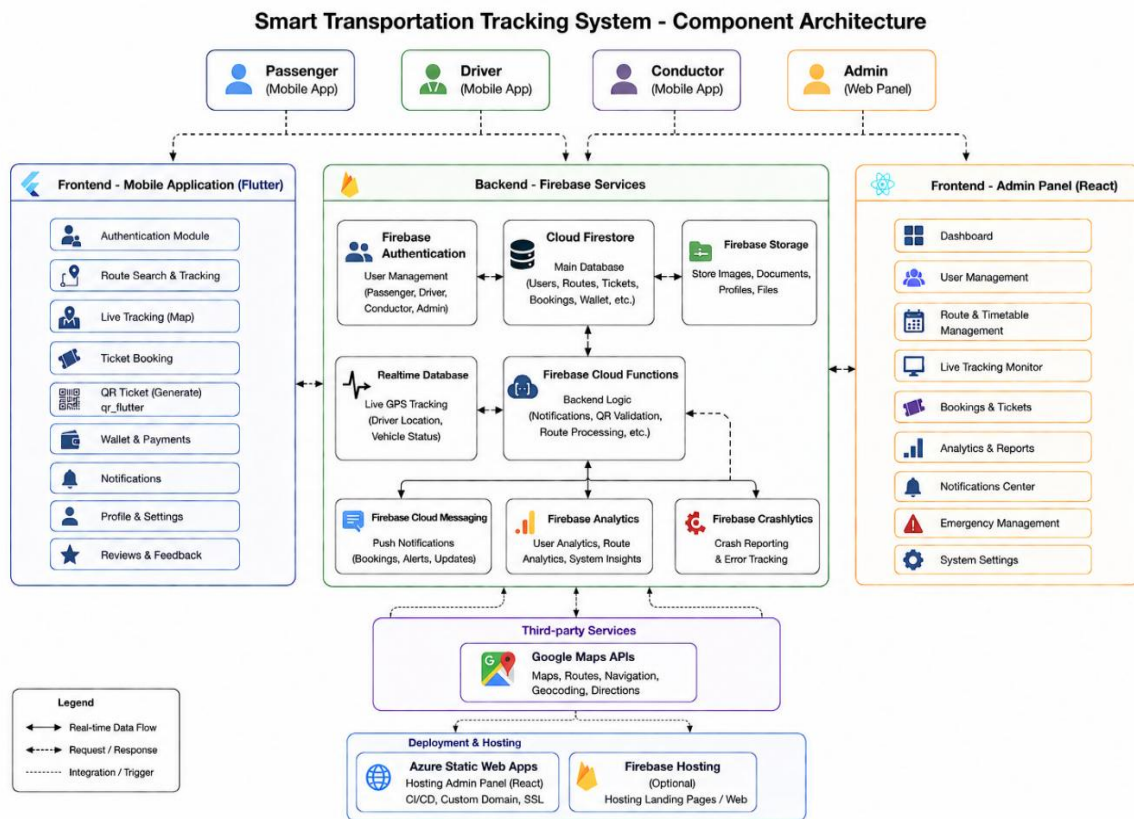


Figure 2: Component Architecture

4. Aim and Objectives

4.1 Aim

The aim of this project is to develop a Smart Public Transportation System for the Badulla District that integrates both bus and train services into a single mobile application, providing real-time tracking, schedules, route information, and transport updates to improve passenger convenience and transportation efficiency.

4.2 Objectives

- To develop a mobile application for public transportation services.
- To provide real-time tracking for buses and transport updates for trains.
- To allow passengers to view routes, schedules, and estimated arrival times.
- To integrate both bus and train services into a single platform.
- To improve passenger convenience and reduce waiting time in public transportation.

5. Methodology

5.1 Software Development Methodology

This project will be developed using the Iterative Software Development Model. The system will be built and improved through multiple iterations, where each iteration focuses on planning, designing, developing, and testing specific features of the application.

Using the iterative model allows the development team to identify issues early, gather feedback continuously, and gradually improve the system. Features such as real-time tracking, route management, and schedule updates can be implemented step by step, ensuring better flexibility and system quality throughout the development process.

5.2 Technology Stack

The proposed Smart Transportation Tracking System uses a modern and scalable technology stack to support real-time transportation management for passengers, drivers, conductors, and administrators. Flutter is used for the mobile application development because it allows the system to run on both Android and iOS platforms using a single codebase, reducing development time and improving performance. React is selected for the admin panel development due to its fast-rendering capabilities and efficient dashboard management features. Firebase Authentication is used to provide secure login and registration functionality for all user roles including passengers, drivers, conductors, and administrators. Cloud Firestore is used as the main synchronized database because it supports real-time data synchronization, flexible document storage, and cloud scalability. Firebase Realtime Database is specifically used for live GPS tracking and vehicle location updates because it handles continuous real-time data transmission more efficiently. Google Maps APIs are integrated into the system to provide route visualization, navigation, location tracking, and estimated travel information for users within the transportation network.

In addition, Firebase Cloud Messaging (FCM) is used to send push notifications such as booking confirmations, emergency alerts, route delays, and travel updates to users in real time. Firebase Cloud Functions act as the backend logic layer of the system by handling server-side operations including automatic notifications, QR validation, route processing, and other background tasks without requiring a dedicated server infrastructure. For digital ticket management, the `qr_flutter` package is used to generate QR-based tickets while the mobile scanner package enables conductors to scan and verify passenger tickets efficiently. Riverpod is used as the state management solution in Flutter to maintain scalable and organized application data flow throughout the system. Firebase Analytics is integrated to monitor user engagement, route popularity, and system performance, helping administrators make data-driven decisions. Firebase Storage is used to securely store images, profile photos, and transportation-related files. The admin panel and web services are deployed using Azure Static Web Apps, which provides reliable cloud hosting and automatic deployment integration. Additionally, Firebase Crashlytics is optionally integrated for professional crash monitoring and error tracking to improve application stability and maintenance.

5.3 Functional Requirement

Passenger Functional Requirements

1. The system shall allow passengers to register and log in securely.
2. The system shall allow passengers to search bus and train routes.
3. The system shall display live vehicle locations using GPS tracking.
4. The system shall allow passengers to book tickets digitally.
5. The system shall generate QR-based tickets after successful booking.
6. The system shall allow passengers to manage their profile information.
7. The system shall display travel history to passengers.
8. The system shall send notifications about delays, bookings, and emergencies.
9. The system shall allow passengers to submit reviews and feedback.

Driver Functional Requirements

1. The system shall allow drivers to log in securely.
 2. The system shall provide drivers with live timetable information.
 3. The system shall display navigation routes using Google Maps.
 4. The system shall update driver location in real time.
 5. The system shall allow drivers to manage profile information.
 6. The system shall maintain driver route history.
 7. The system shall provide driver assistance
1. The system shall allow conductors to log in securely.
 2. The system shall allow conductors to scan and verify QR tickets.
 3. The system shall display booked seat information.
 4. The system shall display seat availability status.
 5. The system shall allow conductors to report accidents or emergencies.
 6. The system shall provide map assistance for conductors.

Admin Functional Requirements

1. The system shall allow administrators to manage users.
2. The system shall allow administrators to add drivers and conductors.
3. The system shall allow administrators to manage transportation timetables.
4. The system shall allow administrators to monitor live driver locations.
5. The system shall allow administrators to manage passenger feedback.
6. The system shall allow administrators to send system notifications.
7. The system shall display booking information and system analytics.
8. The system shall support emergency management operations.

System Functional Requirements

1. The system shall support Android and iOS platforms using Flutter.
2. The system shall support real-time synchronization using Firebase.
3. The system shall integrate Google Maps APIs for navigation and tracking.
4. The system shall provide push notifications using Firebase Cloud Messaging.
5. The system shall store files and images using Firebase Storage.
6. The system shall support QR code generation and scanning.
7. The system shall support analytics features.
8. The system shall maintain secure authentication and authorization.

5.4 Non-Functional Requirements

Performance Requirements

1. The system should update live GPS locations within a few seconds.
2. The system should support multiple concurrent users efficiently.
3. The system should load maps and routes quickly.

Security Requirements

1. User authentication shall be secured using Firebase Authentication.
2. User data shall be encrypted during transmission using HTTPS.
3. Only authorized users shall access restricted features.
4. QR ticket verification shall prevent duplicate ticket usage.

Reliability Requirements

1. The system should provide high availability with minimal downtime.
2. The system should automatically recover from temporary connection failures.
3. Cloud data should be backed up securely.

Scalability Requirements

1. The system should support increasing numbers of passengers and vehicles.
2. Firebase cloud infrastructure should allow scalable real-time operations.

Usability Requirements

1. The application should provide a simple and user-friendly interface.
2. Navigation within the app should be easy for all user roles.
3. The system should support responsive layouts for mobile and web platforms.

Maintainability Requirements

1. The system should use modular architecture for easier maintenance.
2. Riverpod should maintain organized state management.
3. Firebase Cloud Functions should simplify backend maintenance.

Compatibility Requirements

1. The mobile application should support Android and iOS devices.
2. The admin panel should support modern web browsers.
3. The system should integrate with Google Maps services and Firebase services

6. Resource Requirements and Budget

6.1 Hardware Requirements

6.1.1 Passenger Mobile Device Requirements

Minimum Hardware Requirements

1. Processor - Quad-core 1.8 GHz
2. RAM - 3 GB
3. Storage - 2 GB free space
4. GPS Required
5. Camera Required for QR scanning
6. Internet 4G/Wi-Fi
7. Battery 3000 mAh

Recommended Hardware Requirements

1. Processor - Octa-core 2.2 GHz
2. RAM - 6 GB or higher
3. Storage - 5 GB free space
4. GPS - High-accuracy GPS
5. Camera - 8 MP or higher
6. Internet - 5G/Wi-Fi
7. Battery 4500 mAh or higher

6.1.2 Driver Device Requirements.

Recommended Component Requirement

1. Processor - Octa-core
2. RAM - 4–6 GB
3. GPS - High-accuracy GPS
4. Internet - Stable 4G/5G
5. Battery - 5000 mAh
6. Screen 6 inches or larger

6.1.3 Conductor Device Requirements.

1. Camera Required
2. RAM - 3 GB minimum
3. Internet - Stable mobile data
4. Storage - 2 GB free space

6.1.4 Admin System Requirements (Web Dashboard)

1. Minimum Hardware
2. Component Requirement
3. Processor - Intel i3 / Ryzen 3
4. RAM - 4 GB
5. Storage - 10 GB free
6. Internet - Broadband connection
7. Recommended Hardware
8. Component Recommended
9. Processor - Intel i5/i7
10. RAM - 8–16 GB
11. Storage - SSD storage

12. Monitor - Full HD display

6.2 Software Requirements

6.2.1 Mobile Application Software Requirements

1. Operating System - Android 8+ / iOS 13+
2. Flutter Support - Required
3. Google Play Services Required
4. GPS Services Enabled
5. Camera Permission - Enabled
6. Internet Permission – Enabled

6.2.2 Admin Panel Software Requirements

1. Software Requirement
2. Browser - Chrome, Edge, Firefox
3. JavaScript – Enabled

6.2.3 Development Environment Requirements

1. IDE - Android Studio / VS Code
2. SDK - Flutter SDK
3. Database - Firebase
4. Version Control - Git/GitHub
5. API Testing - Postman
6. Cloud & Backend Requirements
7. ServicePurpose
8. Firebase Authentication - Login system
9. Cloud Firestore - Main database
10. Firebase Realtime Database - Live tracking
11. Firebase Cloud Messaging - Notifications
12. Firebase Storage - File storage
13. Google Maps APIs - Navigation & maps
14. Azure Static Web Apps - Hosting
15. Network Requirements
16. GPS Tracking Stable internet
17. Real-time Sync - Low latency
18. Notifications - Internet required
19. QR Verification - Online verification

7. Project Time Plan

Task	Week														
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Requirement Analysis & Research															
System Design & UI Planning															
Iteration 1 – Basic System Development															
Iteration 1 Testing															
Iteration 2 – Bus Module Development															
Iteration 2 Testing															
Iteration 3 – Train & Real-Time Features															
System Integration & Final Testing															
Documentation															

Figure 3: Gantt Chart

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9. Signature of the Team Members

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Date:

Recommendation of the supervisor(s)

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Department/ Organization:

Signature: